End-to-end Handover Simulator for 5G Heterogeneous Network Environments

Introduction

This repository contains a full-stack 5G HetNet Simulator designed to evaluate the performance of various handover algorithms in heterogeneous network environments. The simulator couples:

- A network-level simulator built on ns-3 3.48 with the 5G-LENA module, which generates realistic PHY/MAC traces for each UE-gNB pair.
- A high-level handover simulator written in Python that consumes those traces and runs different handover decision algorithms.

This architecture decouples the physical-layer simulation from the algorithmic evaluation: ns-3 traces are generated once and reused across all handover algorithms.

Tested with:

- ns-3 3.48
- 5G-LENA 5.1

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Architecture Overview

The simulator has two layers:

1. Network simulator (network-simulator/sim.cc, C++/ns-3+5G-LENA) – runs one simulation per UE-gNB pair and writes the results as CSV traces.

2. Handover simulator (handover-simulator/src/main.py, Python) – reads those traces and evaluates the four handover algorithms, producing per-algorithm results.csv files.

See docs/architecture.md for a detailed description of the data flow.

Handover Algorithms

#	Algorithm	Module	Description
1	3GPP Rel.15 A3	simulator_3gpp.py	Standard A3 event-triggered handover with configurable hysteresis and Time-To-Trigger (TTT).
2	3GPP Rel.16 CHO	simulator_3gpp_rel16.py	Conditional Handover preparation and execution are decoupled; execution occurs only when a condition is met.
3	Score-Based Greedy Handover (SBGH)	simulator_sbgh.py	Proposed heuristic that scores candidate cells using RSRP, bandwidth, and carrier frequency, then greedily selects the best cell. Includes an ideal (zero-delay) variant.
4	Multi-Agent DDQN	simulator_gti_dqn.py	Multi-agent Double Deep Q-Network that learns an optimal per-UE handover policy via experience replay.

See docs/algorithms.md for full algorithm details.

Project Structure

```
5g-handover-sim/
├── dockerfile                  # Docker build file
├── docker-compose.yml          # Docker Compose service definition
├── requirements.txt            # Python dependencies (used by Docker)
├── docs/                       # Extended documentation
│   ├── architecture.md
│   ├── algorithms.md
│   ├── configuration.md
│   └── scenario-format.md
├── network-simulator/          # ns-3 C++ simulation module
│   ├── sim.cc                  # Main simulation entry point
│   └── sim.h
└── handover-simulator/         # Python high-level simulator
    ├── requirements.txt
    ├── scenario/
```

```
│   └── sc.txt              # Default scenario definition
├── waypoints/
│   └── wp.txt              # Default UE mobility waypoints
└── src/
    ├── main.py             # Entry point & CLI
    ├── environment.py      # DDQN environment abstraction
    ├── dqn.py              # DQNAgent (DDQN implementation)
    ├── simulator_3gpp.py   # 3GPP Rel.15 algorithm
    ├── simulator_3gpp_rel16.py # 3GPP Rel.16 CHO algorithm
    ├── simulator_sbgh.py   # SBGH algorithm
    ├── simulator_gti_dqn.py # Multi-agent DDQN algorithm
    ├── simulator_common.py # Shared simulation utilities
    ├── scoring.py          # SBGH scoring functions
    ├── utils.py            # General utilities & data loaders
    ├── logging.conf        # Logging configuration
    ├── occupation.py       # gNB load helpers
    └── nrEvents.py         # NR measurement event helpers
```

Prerequisites

Docker (recommended)

- Docker Engine >= 20.10
- Docker Compose >= 2.0

Native

- Ubuntu 24.04 (recommended)
- ns-3 3.48 with 5G-LENA 5.1
- Python 3.10+

Installation

Option A – Docker (recommended)

The Docker image automatically installs ns-3.48, 5G-LENA 5.1, and all Python dependencies.

1. Verify Docker is running:

```
sudo systemctl status docker.service
sudo systemctl status docker.socket
```

2. (Optional) Add your user to the docker group to avoid sudo:

```
sudo usermod -aG docker $USER
newgrp docker
```

3. Build and start the container:

```
docker compose up -d --build
```

The ns-3 build is limited to 4 parallel compilation jobs (NS3_BUILD_JOBS in docker-compose.yml), because the 5G-LENA translation units need roughly 2 GB of memory each and the build is otherwise killed by the OOM handler on Docker installations with a small memory limit. Raise it if your Docker VM has more RAM: docker compose build --build-arg NS3_BUILD_JOBS 8. Do not run two builds of the image at the same time as they will compete for the same memory.

4. Open an interactive shell inside the container:

```
docker compose run simulator
```

5. Continue to Running the Simulator.

Option B – Native

1. Set up ns-3 and 5G-LENA following the official guide. Install ns-3 in the ns-3-dev directory at the root of this repository.

2. Create the symbolic link for the network simulator:

```
ln -s /absolute/path/to/network-simulator /absolute/path/to/ns-3-dev/scratch/network-simulator
```

3. Install Python dependencies:

```
pip3 install -r requirements.txt
```

Running the Simulator

1. Build ns-3 (optimized profile):

```
cd ns-3-dev
./ns3 configure --build-profile=optimized
./ns3 build
```

2. Navigate to the handover simulator:

```
cd ../handover-simulator/src
```

3. Run the simulator:

```
python3 main.py
```

To see all available options:

```
python3 main.py --help
```

4. Re-use existing traces (skip ns-3):

```
python3 main.py --trace /path/to/traces/<run-folder>
```

5. Use a YAML configuration file:

```
python3 main.py --config my_config.yaml
```

Configuration Reference

Argument	Default	Description
--simTime	10.0	Simulation duration in seconds (0 = derive from waypoints)
--nUEs	5	Number of User Equipment nodes
--seed	1234	Random number generator seed
--speed	10.0	UE speed in m/s
--trajectoryTime	5.0	Trajectory segment duration in seconds
--int	0.1	Sampling interval in seconds
--sc	../../handover-simulator/scenario/sc.txt	Scenario definition file
--wp	../../handover-simulator/waypoints/wp.txt	Waypoints file
--trace	(none)	Path to existing trace folder (skips ns-3)
--config	(none)	Path to a YAML configuration file

--Hys	5.0	A3 event hysteresis in dBm
--A3Offset	3.0	A3 event offset in dBm
--ttt	0.1	Time-To-Trigger in seconds
--NrMeasureInt	0.1	NR measurement interval in seconds
--HOInterval	0.1	Handover evaluation interval in seconds
--alpha	5000.0	SBGH scoring weight for bandwidth/RSRP
--beta	1000.0	SBGH secondary scoring parameter
--packetSize	1000	UDP packet size in bytes
--bitRate	380000000	UDP traffic bitrate in bits/s
--MaxPackets	0	Max packets per flow (0 = unlimited)
--errorModel	ns3::NrEesmCcT1	ns-3 error model
--tolerance	1.0	Mobility position tolerance in meters
--logging	0	Enable ns-3 component logging (1 = on)

See docs/configuration.md for full details and examples.

Scenario & Waypoints Files

Scenario file (sc.txt)

Defines the simulation area, frequency bands, and gNodeB placement.

```
# Scenario dimensions (MinX MaxX MinY MaxY)
-400 400 -200 200

# Bands (* Band_ID Central_Freq_Hz User_BW_Hz GNB_BW_Hz)
* 0 3.5e9 20e6 400e6
* 1 28e9 100e6 400e6

# gNodeBs (ID X Y Z Band_ID Tx_Power_dBm Type [I|H])
0 -200.0 50.0 3.0 1 23.0 I # Micro isotropic (mmWave)
6 0.0 150.0 20.0 0 53.0 H # Macro hexagonal (sub-6 GHz)
```

See docs/scenario-format.md for the full file format reference, including the waypoints file syntax.

Output Format

Each run produces a timestamped folder under handover-simulator/src/traces/:

```
traces/<YYYY-MM-DD_HH-MM-SS>/
├── parameters.json              # Simulation parameters snapshot
├── <ue>/<gnb>/traces.csv        # Raw ns-3 traces per UE-gNB pair
└── results/
    ├── 3GPP_A3/results.csv
    ├── 3GPP_CHO/results.csv
    ├── SBGH/results.csv
    ├── ideal-SBGH/results.csv
    └── DDQN/results.csv
```

Each results.csv contains per-interval metrics: throughput, handover count, packet loss, and accumulated bytes.

Documentation

Extended documentation is in the docs/ folder:

Document	Description
docs/architecture.md	System architecture and data flow
docs/algorithms.md	Detailed description of each handover algorithm
docs/configuration.md	Full CLI and YAML configuration reference
docs/scenario-format.md	Scenario and waypoints file format specification